



What makes us tick?

Read the original Nobel Prize release in 2017 for Physiology or Medicine.
<https://digit.in/sep21-23>



Deep brain photoreceptors

Researchers are hunting for deep brain photoreceptors in mammals as well.
<https://digit.in/sep21-24>



INNER CLOCKS

The timekeeping machinery within organisms

Aditya Madanapalle | aditya@digit.in

T

he 2017 Nobel Prize for Physiology and Medicine was awarded to the scientists who

discovered the molecular machinery within animals and plants that kick in at different times of the day. The press release reads, "Jeffrey C. Hall, Michael Rosbash and Michael W. Young were able to peek inside our biological clock and elucidate its inner workings. Their discoveries explain how plants, animals and humans adapt their biological rhythm so that it is synchronized with the Earth's revolutions."

In plants, animals and fungi there are tiny clocks within each cell, regulated by hormones and proteins.

Even the simplest of organisms have internal cellular machinery that takes cues from daylight, such as the bioluminescent cyanobacteria that glow at night, or the plants that switch from night time nitrogen fixation to daytime photosynthesis. In all nonmammalian vertebrates, that includes fish, amphibians, reptiles and birds, there are deep brain photoreceptors connected to the pineal gland, that are then networked to the pituitary gland. In these brains the light can penetrate the skull, allowing for these photoreceptors to provide an additional source of input for the biological clocks. The pineal glands in some nonmammalian species ranging from zebrafish to chickens have oscillators that also provide inputs for these daily rhythms. The machinery within each creature is unique, but the purpose of the internal body clock is the same. In mammals, there is a small portion of the hypothalamus called the Suprachiasmatic nucleus, or SCN, which acts as the central clock responsible for synchronising all the cells in the body, giving rise to

what we know as circadian rhythms. These are natural and daily changes related to temperature regulation, metabolism, hormone release and the sleep cycle, and are maintained by this endogenous clock, taking cues from external changes in the environment such as day-night cycles, temperature variations, timings for meals and social cues. These environmental cues are known as zeitgebers, the German word for "timegivers". Rats kept completely in the dark can maintain the synchronisation between their body clocks and the day night cycles. In rats with damaged SCNs, the amount of time they spent sleeping was the same, but the periodicity and regularity of these sleep timings was destroyed.

What exactly are all the zeitgebers and all the components as well as how the machinery comes together within the body is still a matter of active research, but the secretion of melatonin by the pineal gland is well understood. The SCN suppresses the secretion of melatonin from the pineal gland during the daytime hours, and